

Harrison's Principles of Internal Medicine, 21e >

Chapter 431: Alzheimer's Disease

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ALZHEIMER'S DISEASE

Approximately 50 million people across the world are living with dementia. Alzheimer's disease (AD) is the most common cause of dementia, contributing to an estimated 60–70% of all cases. It is estimated that the median annual total cost of caring for a single patient with advanced AD is >\$50,000, while the emotional toll for family members and caregivers is immeasurable. AD can manifest as early as the third decade of life, but it is the most common cause of dementia in the elderly. Patients most often present with an insidious loss of episodic memory followed by a slowly progressive dementia. In typical amnesic AD, brain atrophy begins in the medial temporal lobes before spreading to the inferior temporal, lateral, medial parietal, and dorsolateral frontal cortices. Microscopically, there are widespread neuritic plaques containing amyloid beta (A β), neurofibrillary tangles (NFTs) composed of hyperphosphorylated tau filaments, and A β accumulation in blood vessel walls in cortex and leptomeninges (amyloid angiopathy, see “Pathology,” below). The identification of causative mutations and susceptibility genes for AD has provided a foundation for progress in understanding the biologic basis of the disorder. The major genetic risk factor for AD is the ϵ 4 allele of the apolipoprotein E (*ApoE*) gene. Carrying one ϵ 4 allele increases the risk for AD by two- to threefold in women whereas carrying two alleles increases the risk ten- to fifteenfold in both sexes. Rapid progress in the development of imaging, cerebrospinal fluid (CSF), and plasma biomarkers of A β and phosphorylated tau has enabled detection of AD pathologic hallmarks in living people, opening the door to early detection and intervention with biologically specific therapies.

CLINICAL MANIFESTATIONS

The cognitive changes of AD tend to follow a characteristic pattern, beginning with memory impairment and progressing to deficits in executive, language, and visuospatial functions. Yet ~20% of patients with AD present with nonmemory complaints such as word-finding, organizational, or navigational difficulty. In other patients, visual processing dysfunction (referred to as posterior cortical atrophy syndrome) or a progressive “logopenic” aphasia characterized by difficulties with naming and repetition are the primary manifestations of AD for years before progressing to involve memory and other cognitive domains. Still, other patients may present with an asymmetric akinetic-rigid-dystonic (“corticobasal”) syndrome or a dysexecutive/behavioral, i.e., “frontal” variant of AD. Depression, social withdrawal, and anxiety occur in early disease stages and may represent a prodrome before cognitive symptoms are apparent.

In early stages of typical amnesic AD, the memory loss may go unrecognized or be ascribed to benign forgetfulness of aging. The term *subjective cognitive decline* refers to self-perceived worsening in memory or other cognitive abilities that may not be noticeable to others or apparent on formal neuropsychologic testing. Once the memory loss becomes noticeable to the patient and family and friends and is confirmed on standardized memory tests, the term *mild cognitive impairment* (MCI) is often used. This construct provides useful prognostic information, because ~50% of patients with MCI (roughly 12% per year) will progress to AD over 4 years. Increasingly, the MCI construct is being replaced by the notion of “early symptomatic AD” to signify that AD is considered the underlying disease (based on clinical or biomarker evidence) in a patient who remains functionally compensated. Even earlier in the course, “preclinical AD” refers to a person with CSF or positron emission tomography (PET) biomarker evidence of amyloid pathology (with or without tau pathology) in the absence of symptoms. It is estimated that preclinical biomarker changes may precede clinical symptoms by 20 years or more, creating a window of opportunity for early-stage treatment and prevention trials. New evidence suggests that partial and sometimes generalized seizures herald AD and can occur even prior to dementia onset, especially in younger patients and patients with autosomal dominant AD-causing mutations.

Eventually, with AD, the cognitive problems begin to interfere with daily activities, such as keeping track of finances, following instructions on the job, driving, shopping, and housekeeping. Some patients are unaware of these difficulties (*anosognosia*), but most remain acutely attuned to their deficits in early disease stages. Changes in environment (travel, relocation, hospitalization) tend to destabilize the patient. Over time, patients become lost on walks or while driving. Social graces, routine behavior, and superficial conversation may be surprisingly intact, even into the later stages of the illness.

In the middle stages of AD, the patient is unable to work, is easily lost and confused, and requires daily supervision. Language becomes impaired—first naming, then comprehension, and finally fluency. Word-finding difficulties and circumlocution can be evident in the early stages, even when formal testing demonstrates intact naming and fluency. *Apraxia* emerges, manifesting as trouble performing learned sequential motor tasks such as using utensils or appliances. Visuospatial deficits begin to interfere with dressing, eating, or even walking, and patients fail to solve simple puzzles or copy geometric figures. Simple calculations and clock reading become difficult in parallel.

In the late stages, some persons remain ambulatory, wandering aimlessly. Loss of judgment and reasoning is inevitable. Delusions are prevalent and usually simple, with common themes of theft, infidelity, or misidentification. Disinhibition and uncharacteristic belligerence may occur and alternate with passivity and withdrawal. Sleep-wake patterns are disrupted, and nighttime wandering becomes disturbing to the household. Some patients develop a shuffling gait with generalized muscle rigidity associated with slowness and awkwardness of movement. Patients often look parkinsonian (**Chap. 435**) but rarely have a high-amplitude, low-frequency tremor at rest. There is a strong overlap between dementia with Lewy bodies (DLB) (**Chap. 434**) and AD, and some AD patients develop more classical parkinsonian features.

In the end stages, patients with AD become rigid, mute, incontinent, and bedridden, and need help with eating, dressing, and toileting. Hyperactive tendon reflexes and myoclonic jerks (sudden brief contractions of various muscles or the whole body) may occur spontaneously or in response to physical or auditory stimulation. Often death results from malnutrition, secondary infections, pulmonary emboli, heart disease, or, most commonly, aspiration. The typical duration of symptomatic AD is 8–10 years, but the course ranges from 1–25 years. For unknown reasons, some patients with AD show a steady decline in function while others have prolonged plateaus without major deterioration.

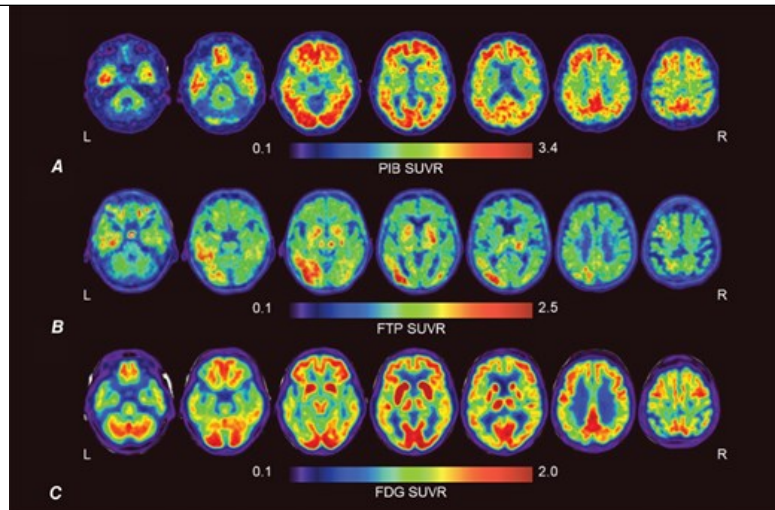
DIAGNOSIS

A detailed discussion of the diagnosis of dementia is presented in **Chap. 29**. Early in the disease course, other etiologies of dementia should be excluded (see **Tables 29-1, 29-3, and 29-4**). Neuroimaging studies (CT and MRI) do not show a single specific pattern with AD and may be normal early in the disease. As AD progresses, more distributed but usually posterior-predominant cortical atrophy becomes apparent, along with atrophy of the medial temporal memory structures (see **Fig. 29-1**). The main purpose of imaging is to exclude other disorders, such as primary and secondary neoplasms, vascular dementia, diffuse white matter disease, and normal-pressure hydrocephalus (NPH). Imaging also helps to distinguish AD from other degenerative disorders, such as frontotemporal dementia (FTD) (**Chap. 432**) or the prion disorder Creutzfeldt-Jakob disease (CJD) (**Chap. 438**), which feature imaging patterns that are different from AD. Functional imaging studies, such as fluorodeoxyglucose (FDG) PET, reveal hypometabolism in the posterior temporal-parietal cortex in AD (see **Fig. 29-1**).

Amyloid PET imaging (e.g., with radiotracers [^{11}C]PIB, [^{18}F]florbetapir, [^{18}F]florbetaben, or [^{18}F]flutemetamol) confirms the presence of neuritic and diffuse A β plaques throughout the neocortex (**Fig. 431-1**). Although amyloid PET binding is detected in AD, approximately 25% of cognitively unimpaired older individuals also have positive scans, thought to represent preclinical disease and an increase in the risk of converting to clinical AD. Similarly, dementia due to a non-AD disorder can be the underlying etiology in a patient who tests positively on amyloid PET. Amyloid PET ligands also bind to vascular A β deposits in cerebral amyloid angiopathy (**Chap. 428**). Therefore, clinical use of amyloid PET should be restricted to specific scenarios in which knowledge of amyloid status is expected to impact diagnosis and change management. For example, a negative amyloid PET scan in a patient with dementia makes an AD diagnosis unlikely.

FIGURE 431-1

Molecular imaging of Alzheimer’s disease pathophysiology in an 81-year-old with mild Alzheimer’s disease. A. A β positron emission tomography (PET) with [^{11}C]PIB reveals extensive radiotracer retention in neocortex, consistent with the known distribution of amyloid plaques. **B.** Tau PET with [^{18}F]FTP shows asymmetric uptake predominantly in left temporal cortex, consistent with intermediate-stage NFTs. Tracer uptake in midbrain and basal ganglia represents “off-target” (non-tau related) tracer retention. **C.** FDG-PET reveals reduced tracer uptake in left greater than right temporal and parietal cortex, indicative of decreased synaptic activity. The pattern of hypometabolism corresponds more closely to the pattern of tau than amyloid deposition. **A–C.** Axial brain slices are shown in neurologic orientation. L, left; R, right; SUVR, standardized uptake value ratio, a quantitative measure of PET radiotracer retention.



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Tau PET radiotracers (e.g., [¹⁸F]flortaucipir, [¹⁸F]MK-6240) bind to the paired helical filaments that form NFTs and are primarily available in the research setting. The pattern of binding is largely consistent with Braak neuropathologic staging of NFTs, with early retention in medial temporal regions, followed by spread into temporoparietal and cingulate cortices, dorsolateral prefrontal regions, and, ultimately, primary sensory and motor areas.

Routine spinal fluid examination is generally normal, but CSF reductions in Aβ₄₂ levels and the Aβ₄₂/Aβ₄₀ ratio correlate with amyloid deposition, increases in phosphorylated tau (at residue 181 or 217) correlate with tangle inclusions, and increases in total tau levels represent a nonspecific finding seen in AD but also in other causes of neurodegeneration. Plasma measurements of Aβ and phosphorylated tau with ultra-sensitive immunoassays or mass spectrometry show great promise and are likely to increase access and affordability of AD molecular biomarkers.

Electroencephalogram (EEG) is normal or shows nonspecific slowing; prolonged EEG can be used to seek out intermittent nonconvulsive seizures.

Slowly progressive decline in memory and orientation, normal results on laboratory tests, and an MRI or CT scan showing only distributed or posteriorly predominant cortical and hippocampal atrophy are highly suggestive of AD. A clinical diagnosis of AD reached after careful evaluation is confirmed at autopsy 70–90% of the time, with misdiagnosed cases usually resulting from pathologic limbic-predominant age-related TDP-43 encephalopathy (LATE) neuropathologic changes with or without hippocampal sclerosis, primary age-related tauopathy, DLB, vascular pathology, or frontotemporal lobar degeneration (FTD).

Simple clinical clues are useful in the differential diagnosis. Early prominent gait disturbance with only mild memory loss suggests vascular dementia or, rarely, NPH (see below). Resting tremor with stooped posture, bradykinesia, and masked facies suggest PD ([Chap. 435](#)) or DLB ([Chap. 434](#)). When dementia occurs after a well-established diagnosis of PD, PD dementia (PDD) is usually the correct diagnosis, but many patients with this diagnosis will show a mixture of AD and DLB at autopsy. The early appearance of parkinsonian features in association with fluctuating alertness, visual hallucinations, or delusional misidentification suggests DLB. Chronic alcoholism should prompt the search for vitamin deficiency. Loss of joint position and vibration sensibility accompanied by Babinski signs suggests vitamin B₁₂ deficiency, especially in a patient with a history of autoimmune disease, small bowel resection or irradiation, or veganism ([Chap. 99](#)). Early onset of a focal seizure suggests a metastatic or primary brain neoplasm ([Chap. 90](#)). Previous or ongoing depression raises suspicion for depression-related cognitive impairment, although significant cognitive changes with depression are uncommon and AD and DLB can feature a depressive or anxious prodrome. A history of treatment for insomnia, anxiety, psychiatric disturbance, or epilepsy suggests chronic drug intoxication. Rapid progression over a few weeks or months associated with rigidity and myoclonus suggests CJD ([Chap. 438](#)). Prominent behavioral changes with intact navigation and focal anterior-predominant atrophy on brain imaging are typical of FTD. A positive family history of dementia suggests either one of the familial forms of AD or one of the other genetic disorders associated with dementia, such as FTD ([Chap. 432](#)), Huntington's disease (HD) ([Chap. 436](#)), prion disease ([Chap. 438](#)), or rare hereditary ataxias ([Chap. 439](#)).

EPIDEMIOLOGY

The most important risk factors for AD are increasing age and a positive family history. In the United States, approximately 10% of people over age 65

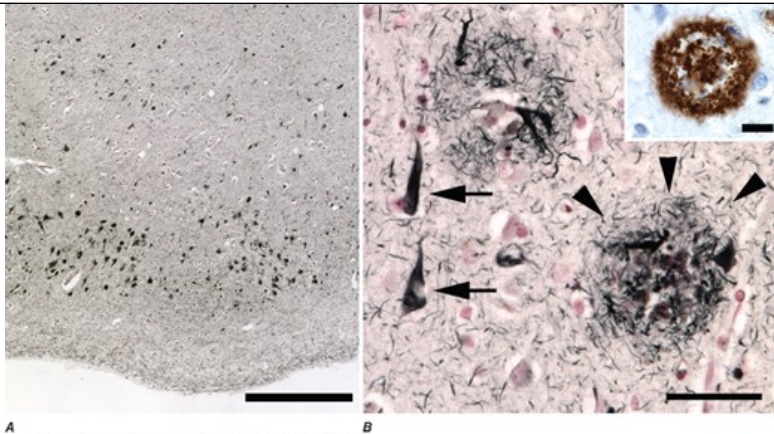
years have AD, including 3% of people age 65–74 years, 17% of people age 75–84 years, and 32% of people age 85 years and older. A positive family history of dementia suggests a genetic contribution to AD, which is usually attributable to the *ApoE ε4* risk allele. Autosomal dominant inheritance occurs in only 1–2% of patients and is typically accompanied by a multigenerational history of early-onset dementia. Female sex is a risk factor independent of the greater longevity of females, and women who carry a single *ApoE ε4* allele are more susceptible than are male *ε4* carriers. A history of mild-to-severe traumatic brain injury increases the risk for AD. AD is more common in groups with low educational attainment, but education influences test-taking ability, and it is clear that AD can affect persons of all intellectual levels. One study found that the capacity to express complex written language in early adulthood correlated with a decreased risk for AD. Similarly, illiteracy and low educational attainment are risk factors for dementia. Numerous environmental factors, including aluminum, mercury, and viruses, have been proposed as causes of AD, but rigorous studies have failed to demonstrate a significant role for any of these exposures. Similarly, several studies suggest that the use of nonsteroidal anti-inflammatory agents is associated with a decreased risk of AD, but this risk has not been confirmed in large prospective studies. Vascular disease, and stroke in particular, seems to lower the threshold for the clinical expression of AD. Also, in many patients with AD, amyloid angiopathy can lead to microhemorrhages, large lobar hemorrhages, ischemic infarctions most often in the subcortical white matter, or in rare cases an inflammatory leukoencephalopathy. Diabetes increases the risk of AD threefold. Elevated homocysteine and cholesterol levels; hypertension; obesity; hearing loss; tobacco use; diminished serum levels of folic acid; low dietary intake of fruits, vegetables, and red wine; sleep disorders; low levels of exercise; and air pollution exposure are all being explored as potential risk factors for dementia in general and AD in particular.

PATHOLOGY

At autopsy, the earliest and most severe degeneration is usually found in the medial temporal lobe (entorhinal/perirhinal cortex and hippocampus), inferolateral temporal cortex, and nucleus basalis of Meynert. The characteristic microscopic findings are neuritic plaques and NFTs (**Fig. 431-2**). These lesions accumulate in small numbers during normal brain aging but dominate the picture in AD. The overall burden of AD neuropathologic changes can be graded based on the topography of Aβ plaques, the density of neuritic plaques, and the spatial extent of NFTs present. Increasing evidence suggests that soluble amyloid species called *oligomers* may cause cellular dysfunction and represent the early toxic molecule in AD. Eventually, further amyloid polymerization and fibril formation lead to neuritic plaques, which contain a central core of amyloid, proteoglycans, ApoE, α-antichymotrypsin, and other proteins. Aβ is a protein of 39–42 amino acids that is derived proteolytically from a larger transmembrane protein, *amyloid precursor protein* (APP), when APP is cleaved by β and γ secretases (**Fig. 431-3**). The normal function of the Aβ peptides remains uncertain. APP has neurotrophic and neuroprotective properties. The plaque core is surrounded by a halo, which contains dystrophic, tau-immunoreactive neurites and activated microglia. The accumulation of Aβ in cerebral arterioles is termed *amyloid angiopathy*. NFTs are composed of silver-staining neuronal cytoplasmic fibrils composed of abnormally phosphorylated tau protein; they appear as paired helical filaments by electron microscopy. Tau binds to and stabilizes microtubules, supporting axonal transport of organelles, glycoproteins, neurotransmitters, and other important cargoes throughout the neuron. Once hyperphosphorylated, tau can no longer bind properly to microtubules and redistributes from the axon to throughout the neuronal cytoplasm and distal dendrites, compromising function. Other theories emphasize that abnormal conformations of tau induce misfolding of native (unfolded) tau into pathologic conformations and that this prion-like templating process is responsible for tau spreading (**Chap. 424**). Finally, patients with AD often show comorbid DLB, TDP-43, or vascular pathology. Most prevailing rodent models of AD involve expression of mutant transgenes that leads to Aβ₄₂ accumulation in the absence of tauopathy. Even in these models, diminishing neuronal tau ameliorates cognitive deficits and nonconvulsive seizures while Aβ₄₂ continues to accumulate, raising hope for tau-lowering therapies in humans. Biochemically, AD is associated with a decrease in the cortical levels of several proteins and neurotransmitters, especially **acetylcholine**, its synthetic enzyme choline acetyltransferase, and nicotinic cholinergic receptors. Reduction of **acetylcholine** reflects degeneration of cholinergic neurons in the nucleus basalis of Meynert, located just below the thalamus and adjacent to the third ventricle, that project throughout the cortex. There is also noradrenergic and serotonergic depletion due to degeneration of upper brainstem nuclei such as the locus coeruleus (**norepinephrine**) and dorsal raphe (serotonin), where tau-immunoreactive neuronal cytoplasmic inclusions can be identified in early adult life, even in individuals lacking entorhinal cortex NFTs.

FIGURE 431-2

Neuropathology of Alzheimer's disease. A. Early neurofibrillary degeneration, consisting of NFTs and neuropil threads, preferentially affects the medial temporal lobes, especially the stellate pyramidal neurons that compose the layer 2 islands of entorhinal cortex, as shown using Gallyas **silver** staining. **B.** Higher magnification view reveals the fibrillar nature of tangles (*arrows*) and the complex structure of neuritic plaques (*arrowheads*), whose major component is Aβ (*inset* shows immunohistochemistry for Aβ). Scale bars are 500 μm in **A**, 50 μm in **B**, and 20 μm in **B inset**.

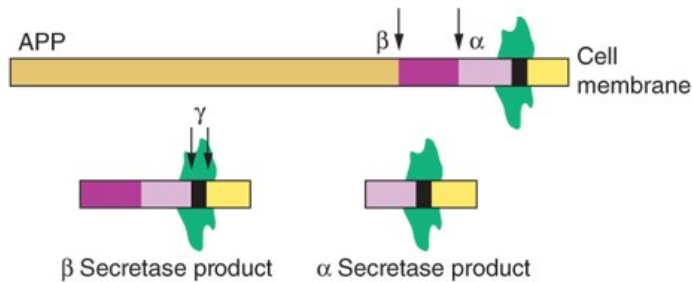


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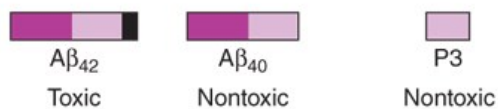
FIGURE 431-3

Amyloid precursor protein (APP) is catabolized by α , β , and γ secretases. A key initial step is the digestion by either β secretase (BACE) or α secretase (ADAM10 or ADAM17 [TACE]), producing smaller nontoxic products. Cleavage of the β secretase product by γ secretase (Step 2) results in either the toxic $A\beta_{42}$ or the nontoxic $A\beta_{40}$ peptide; cleavage of the α secretase product by γ secretase produces the nontoxic P3 peptide. Excess production of $A\beta_{42}$ is a key initiator of cellular damage in Alzheimer's disease (AD). Therapeutics for AD have focused on attempts to reduce accumulation of $A\beta_{42}$ by antagonizing β or γ secretases, promoting α secretase, or clearing $A\beta_{42}$ that has already formed by use of specific antibodies.

Step 1: Cleavage by either α or β secretase



Step 2: Cleavage by γ secretase



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GENETIC CONSIDERATIONS



Several genes play an important role in the pathogenesis of AD. One is the *APP* gene on chromosome 21. Adults with trisomy 21 (Down's syndrome) consistently develop the typical neuropathologic hallmarks of AD if they survive beyond age 40 years, and many develop a progressive dementia superimposed on their baseline deficits. The extra dose of the *APP* gene on chromosome 21 is the initiating cause of AD in adult Down's syndrome and results in excess cerebral amyloid production. Supporting this hypothesis, some families with early-age-of-onset familial AD (FAD) have point mutations in *APP*. Although very rare, these families were the first examples of single-gene autosomal dominant transmission of AD.

Investigation of large families with multigenerational FAD led to the discovery of two additional AD-causing genes, the *presenilins*. Presenilin-1 (*PSEN-1*) is on chromosome 14 and encodes presenilin-1 protein (also known as S182). Mutations in this gene cause an early-age-of-onset AD, with onset

typically before age 60 and often before age 50, transmitted in an autosomal dominant, highly penetrant fashion. More than 100 different mutations have been found in the *PSEN-1* gene in families from a wide range of ethnic backgrounds. Presenilin-2 (*PSEN-2*) is on chromosome 1 and encodes the presenilin-2 protein (also known as STM2). A mutation in the *PSEN-2* gene was first found in a group of American families with Volga German ethnic background. Mutations in *PSEN-1* are much more common than those in *PSEN-2*. The presenilins are highly homologous and encode similar proteins that at first appeared to have seven transmembrane domains (hence the designation *STM*), but subsequent studies have suggested eight such domains, with a ninth submembrane region. Both presenilins are cytoplasmic neuronal proteins that are widely expressed throughout the nervous system. They are homologous to a cell-trafficking protein, sel 12, found in the nematode *Caenorhabditis elegans*. Prior to symptom onset, patients with mutations in the presenilin genes have elevated CSF levels of A β ₄₂, and *PSEN-1* mutations produce increased A β ₄₂ in the media in cell culture.

PSEN-1 is involved in the cleavage of APP at the γ secretase site and mutations in either gene (*PSEN-1* or *APP*) may disturb γ secretase cleavage. Mutations in *PSEN-1* are the most common cause of early-age-of-onset FAD, representing 40–70% of all cases. Mutations in *PSEN-1* tend to produce AD with an earlier age of onset (mean onset 45 years) and a shorter, more rapidly progressive course (mean duration 6–7 years) than mutations in *PSEN-2* (mean onset 53 years; duration 11 years). Although some carriers of *PSEN-2* mutations have had onset of dementia after the age of 70 years, mutations in the presenilins rarely lead to late-age-of-onset AD. Clinical genetic testing for these uncommon mutations is available but likely to be revealing only in early-age-of-onset FAD and should be performed in association with formal genetic counseling.

The *APOE* gene on chromosome 19 is involved in the pathogenesis of AD. The protein product, ApoE, participates in cholesterol transport (**Chap. 407**), and the gene has three alleles: ϵ 2, ϵ 3, and ϵ 4. The *ApoE* ϵ 4 allele confers increased risk of AD in the general population, including sporadic and late-age-of-onset familial forms. Approximately 24–30% of the nondemented white population has at least one ϵ 4 allele (12–15% allele frequency), and about 2% are ϵ 4/ ϵ 4 homozygotes. Among patients with AD, 40–65% have at least one ϵ 4 allele, a highly significant elevation compared with controls. The increased risk associated with a single ϵ 4 allele is especially prominent in women. The risk of AD in *ApoE* ϵ 4 carriers also varies by racial and ethnic background, with increased risk in East Asians and decreased risk in blacks and Hispanics compared with whites. Additionally, many patients with AD have no ϵ 4 allele, and ϵ 4 carriers may never develop AD. Therefore, ϵ 4 is neither necessary nor sufficient to cause AD. Nevertheless, the *ApoE* ϵ 4 allele represents the most important genetic risk factor for sporadic AD and acts as a dose-dependent disease modifier, with each *ApoE* ϵ 4 allele associated with an approximately 10-year earlier age of onset. The association between *ApoE* ϵ 4 and AD is strongest in patients 60–85 years of age and is weaker in younger patients and in the very old. The precise mechanisms through which *ApoE* ϵ 4 confers AD risk or hastens onset remain unclear, but ϵ 4 leads to less efficient amyloid clearance and production of toxic fragments from cleavage of the molecule. ApoE can be identified in neuritic plaques and may also be involved in NFT formation, because it binds to tau protein. ApoE ϵ 4 decreases neurite outgrowth in dorsal root ganglion neuronal cultures, perhaps indicating a deleterious role in the brain's response to injury. Increasing evidence suggests that the ϵ 2 allele may reduce AD risk. Use of *ApoE* testing in AD diagnosis remains controversial because its predictive value remains unclear and many individuals with the ϵ 4 allele will never develop dementia. Cognitively normal ϵ 4 heterozygotes and homozygotes may show decreased posterior cerebral cortical metabolic function by PET imaging, suggesting presymptomatic abnormalities due to AD or an inherited vulnerability of the AD-targeted network. In demented persons who meet clinical criteria for AD, finding an ϵ 4 allele increases the reliability of diagnosis; however, the absence of an ϵ 4 allele cannot be considered evidence against AD. Nevertheless, ApoE ϵ 4 remains the single most important biologic marker associated with AD risk, and studies of ϵ 4's functional role and diagnostic utility are progressing rapidly. *ApoE* genotyping is available in some straight-to-consumer genetic testing platforms. The ϵ 4 allele is associated with increased risk for cerebral amyloid angiopathy (CAA), DLB, and vascular dementia, while its association with FTD is uncertain. Some evidence suggests that ϵ 4 may worsen the expression of non-AD degenerative disorders, head trauma, and other brain injuries. Additional genes are also likely to be involved in AD, especially as minor-risk alleles for sporadic forms of the disease. Genome-wide association studies have identified more than 20 additional common genetic variants that, individually, have small (i.e., odds ratios ~1.1–1.2 or 0.8–0.9) impacts on the risk of AD. Implicated genes converge in biologic pathways related to innate immunity, lipid metabolism, and synaptic function. Examples include the clusterin (*CLU*), phosphatidylinositol-binding clathrin assembly protein (*PICALM*), and complement component (3b/4b) receptor 1 (*CR1*) genes, among others. *CLU* may play a role in synapse turnover, *PICALM* participates in clathrin-mediated endocytosis, and *CR1* may be involved in amyloid clearance or synapse loss through the complement pathway. *TREM2* is a gene involved with inflammation that increases the likelihood of dementia. Homozygous mutation carriers develop a frontal dementia with bone cysts (Nasu-Hakola disease), whereas heterozygotes are predisposed to the development of AD. *TREM2* risk alleles are rare but have strong effects, with odds ratios estimated at 3–4 for developing clinical AD. Polygenic hazard scores that integrate the presence of multiple risk and protective alleles may be useful in predicting an individual's lifetime risk of developing AD. The vast majority of AD genetic studies have focused on white populations of European descent, and much less is known about the genetics of AD in nonwhite populations.

TREATMENT

Alzheimer's Disease

The management of AD is challenging and gratifying despite the absence of a cure or a robust pharmacologic treatment. The primary focus is on long-term amelioration of associated behavioral and neurologic problems, as well as providing caregiver support, though many potential disease-modifying therapies are currently being tested in human trials.

PATIENT AND CAREGIVER EDUCATION

Building rapport with the patient, family members, and other caregivers is essential. In the early stages of AD, memory aids such as notebooks and posted daily reminders can be helpful. Family members should emphasize activities that are pleasant while curtailing those that increase stress on the patient. Kitchens, bathrooms, stairways, and bedrooms need to be made safe, and eventually patients will need to stop driving. Loss of independence and change of environment may worsen confusion, agitation, and anger. Communication and repeated calm reassurance are necessary. Caregiver “burnout” is common, often resulting in nursing home placement of the patient or new health problems for the caregiver. Respite breaks for the caregiver help to maintain a successful long-term therapeutic milieu. Use of adult day-care centers can be helpful. Local and national support groups, such as the Alzheimer's Association and the Family Caregiver Alliance, are valuable resources. Internet access to these resources has become available to clinicians and families in recent years.

NEUROTRANSMITTER-BASED THERAPIES

Donepezil (target dose, 10 mg daily), **rivastigmine** (target dose, 6 mg twice daily or 9.5-mg patch daily), **galantamine** (target dose 24 mg daily, extended-release), and **memantine** (target dose, 10 mg twice daily) are approved by the U.S. Food and Drug Administration (FDA) for the treatment of AD. Due to hepatotoxicity, tacrine is no longer used. Dose escalations for each of these medications must be carried out over 4–6 weeks to minimize side effects. The pharmacologic action of **donepezil**, **rivastigmine**, and **galantamine** is inhibition of the cholinesterases, primarily acetylcholinesterase, with a resulting increase in cerebral **acetylcholine** levels. **Memantine** appears to act by blocking overexcited *N*-methyl-D-aspartate (NMDA) glutamate receptors. Double-blind, placebo-controlled, crossover studies with cholinesterase inhibitors (in mild-to-severe AD dementia) and **memantine** (in moderate-to-severe AD dementia) have shown them to be associated with modestly improved caregiver ratings of patients' functioning and with an apparent decreased rate of decline in cognitive test scores over periods of up to 3 years. The average patient on an anticholinesterase inhibitor maintains his or her mini-mental state examination (MMSE) score for close to a year, whereas a placebo-treated patient declines 2–3 points over the same time period. **Memantine**, used in conjunction with cholinesterase inhibitors or by itself, slows cognitive deterioration and decreases caregiver burden for patients with moderate-to-severe AD but is not approved for mild AD. Neither cholinesterase inhibitors nor **memantine** has proven efficacious in patients with MCI. Cholinesterase inhibitors are relatively easy to administer, and their major side effects are gastrointestinal symptoms (nausea, diarrhea, cramps), altered sleep with unpleasant or vivid dreams, bradycardia (usually benign), and muscle cramps. Potential side effects associated with **memantine** include constipation, dizziness, headache, and somnolence. A common approach to AD drug therapy is to initiate a cholinesterase inhibitor for a patient diagnosed with mild AD dementia, and to add **memantine** when patients enter the moderate stage of disease. Cholinesterase inhibitors may also be effective in treating delusions and hallucinations, while **memantine** can reduce agitation.

THERAPIES TARGETING AMYLOID- β

AD drug development over the past two decades has focused on the prevention or clearance of A β pathology. In June 2021, aducanumab, a monoclonal antibody targeting the N-terminus of the A β peptide, was granted accelerated approval by the FDA based on reduction in A β plaques (measured by PET) in two phase 3, double-blinded, randomized placebo-controlled trials. However, a clinical benefit over placebo (measured by slower decline on cognitive and functional scales) was observed with high-dose treatment (10 mg/kg) in only one of the two trials, and lower antibody doses did not show a benefit vs. placebo in either trial. Data interpretation was further complicated by differences in dosing between trials and early termination of both trials based on a prespecified futility analysis, which ultimately proved erroneous. Given these circumstances, continued FDA approval will be contingent upon verification of clinical benefit in confirmatory trials.

According to the FDA label, aducanumab should only be considered for treatment of patients with MCI or early dementia due to AD, mirroring the early clinical stage of patients enrolled in the phase 3 clinical trials. Patients with pre-clinical (i.e., asymptomatic, biomarker-positive) AD or patients with moderate-severe AD dementia should not be treated until data on safety or efficacy in these populations is available. Expert recommendations further stipulate that biomarker confirmation of A β based on CSF or PET be required prior to initiating treatment, since clinical diagnosis in isolation is not sufficient to ensure the presence of A β plaques. Patients with confounding neurological or psychiatric conditions, unstable medical illnesses, evidence of prior brain hemorrhages (including multiple microhemorrhages), or active anticoagulant treatment should be excluded.

Aducanumab is administered as an intravenous infusion every 4 weeks, with gradual dose titration from 1 mg/kg to 10 mg/kg over seven infusions. Amyloid-related imaging abnormalities (ARIA) are the most common adverse effect, occurring in 41% of patients treated with high-dose aducanumab vs. 10% in the placebo groups. ARIA can manifest as vasogenic edema (ARIA-E) or cortical microhemorrhages and superficial siderosis (ARIA-H). Of ARIA cases in the phase 3 trials, 74% were asymptomatic and detected by safety MRIs. The most common symptoms associated with ARIA were headache, altered mental status, dizziness, visual disturbances, and nausea. Symptoms were usually mild and transient, though severe cases with focal neurologic deficits have been described. Most cases occurred within the first eight infusions, though ARIA can occur at any time. Baseline and safety surveillance MRI scans (at minimum following the 7th and 12th infusions) are required for aducanumab treatment, and an MRI is also indicated in treated patients in whom ARIA is suspected clinically. Therefore, patients with contraindications to MRI cannot safely receive this therapy. *ApoE* genotyping may also be considered to inform risk-benefit discussions prior to treatment, because ARIA-E is more common in *ApoE* $\epsilon 4$ carriers (43%) compared to noncarriers (20%). It is highly recommended that the antibody only be prescribed by clinicians who have adequate training and access to the resources needed to safely deliver this complex therapy.

ADDITIONAL THERAPIES

Mild-to-moderate depression is common in the early stages of AD and may respond to antidepressants or cholinesterase inhibitors. Selective serotonin reuptake inhibitors (SSRIs) are commonly used due to their low anticholinergic side effects (for example, [escitalopram](#), target dose 5–10 mg daily). Seizures can be treated with [levetiracetam](#) unless the patient had a different regimen that was effective prior to the onset of AD. Agitation, insomnia, hallucinations, and belligerence are especially troublesome characteristics of some AD patients, and these behaviors can lead to nursing home placement. The newer generation of atypical antipsychotics, such as [risperidone](#), [quetiapine](#), and [olanzapine](#), are being used in low doses to treat these neuropsychiatric symptoms. The few controlled studies comparing drugs against behavioral intervention in the treatment of agitation suggest mild efficacy with significant side effects related to sleep, gait, and cardiovascular complications, including an increased risk of death. All antipsychotics carry a black box FDA warning for use in elderly patients with dementia and thus should be prescribed only with caution; however, careful, daily, nonpharmacologic behavior management is often not available, rendering medications necessary for some patients. Medications with strong anticholinergic effects should be vigilantly avoided, including prescription and over-the-counter sleep aids (e.g., [diphenhydramine](#)) or incontinence therapies (e.g., [oxybutynin](#)).

Several commonly used medications and supplements, including estrogen hormone replacement therapy, statins, vitamin E, and ginkgo biloba, appeared to be associated with a decreased risk of AD in epidemiologic or observational studies, but did not show efficacy in prospective, randomized, double-blinded, placebo-controlled trials. Many [vitamins](#) and dietary supplements are marketed directly to consumers as “memory enhancing” or protective against AD without clinical evidence. Patients and families may come across anecdotal reports of “miraculous” responses to aggressive treatments such as anti-interferon intrathecal infusions, intravenous immunoglobulin, antibiotics (purportedly to treat Lyme disease or another questionable infection), metal chelation, and stem cell therapies, but there is no scientific evidence to support use of any of these approaches to treating AD, and significant concern for harm.

EXPERIMENTAL THERAPIES

The design of AD clinical trials has been transformed by the availability of PET and CSF biomarkers of A β and tau. Many trials now require biomarker evidence of AD for trial inclusion. Biomarkers help assess target engagement (e.g., changes in CSF or PET A β in an anti-amyloid trial) or modification of downstream disease pathophysiology (e.g., changes in CSF or PET tau in an anti-amyloid trial), with the pivotal trials leading to approval of aducanumab being emblematic of this novel approach. Increasingly, many trials have shifted toward enrolling patients in the asymptomatic (preclinical) or very early symptomatic stages of AD, using positive biomarkers as the primary inclusion criterion. Primary (biomarker-negative) and secondary (biomarker-positive but no symptoms) prevention trials are underway in autosomal dominant mutation carriers, *ApoE* $\epsilon 4$ homozygotes, and even in the normally aging population.

Beyond aducanumab, several additional anti-A β monoclonal antibodies (e.g., lecanemab, gantenerumab, and donanemab) have shown evidence of robust amyloid plaque lowering on PET and are currently being evaluated in clinical trials across the continuum from preclinical disease to mild dementia due to AD. As with aducanumab, ARIA-E and ARIA-H represent a safety concern for this class of drugs. Active vaccination against A β is another approach that aims to promote immune-mediated clearance of amyloid pathology. The first A β_{42} vaccine trial in humans was aborted after a minority of patients developed meningoencephalitis, but subsequent trials with less immunogenic formulations have shown more favorable safety profiles.

Oral drugs that inhibit β and γ secretase reduce the cleavage of APP to A β_{42} and showed promise in ameliorating pathology and behavioral changes

in AD transgenic mice. Unfortunately, placebo-controlled trials failed to show clinical efficacy, and trials of β secretase inhibitors in particular, consistently found significant *worsening* of cognition in treated patients vs. placebo, though fortunately this effect proved transient after discontinuing the drug. It is unclear whether toxicity of β and γ secretase inhibitors was directly related to changes in A β metabolism or to “off-target” drug effects.

Monoclonal antibodies directed against phosphorylated tau are in earlier stages of development. These antibodies aim to prevent the transsynaptic spread of tau and have proven effective in tau-transgenic mice. Safety profiles in human studies have proven favorable thus far. Additional therapeutic approaches targeting tau include: active immunization; inhibition of tau phosphorylation, acetylation, and aggregation; microtubule stabilization; and lowering of tau expression via antisense oligonucleotides or small interfering RNA. Other druggable pathways represented in the AD drug development pipeline include neuroinflammation, metabolism/bioenergetics, synaptic plasticity, neuroprotection, and neurotransmitter-based treatment of neuropsychiatric symptoms.

A general approach to the symptomatic management of dementia is presented in [Chap. 25](#).

OTHER CAUSES OF DEMENTIA

FTD ([Chap. 432](#)), vascular dementia ([Chap. 433](#)), DLB ([Chap. 434](#)), and prion diseases ([Chap. 438](#)) are covered in dedicated chapters.

Prion diseases such as CJD are rare neurodegenerative conditions (prevalence ~1 per million) that produce dementia. CJD is a rapidly progressive disorder associated with dementia, focal cortical signs, rigidity, and myoclonus, causing death <1 year after first symptoms appear. The rapidity of progression seen with CJD is uncommon in AD so that the distinction between the two disorders is usually straightforward, although AD can on occasion present as a rapidly progressive dementia. In general, corticobasal degeneration (CBD) ([Chap. 432](#)) and DLB ([Chap. 426](#)), more rapid degenerative dementias with prominent movement abnormalities, are more likely to be mistaken for CJD. The differential diagnosis for CJD includes other rapidly progressive dementing conditions such as viral or bacterial encephalitides, Hashimoto’s encephalopathy, central nervous system (CNS) vasculitis, lymphoma, or paraneoplastic/autoimmune syndromes ([Chap. 94](#)). The markedly abnormal periodic complexes on EEG and cortical ribboning and basal ganglia hyperintensities on diffusion-weighted imaging or fluid-attenuated inversion recovery MRI are diagnostic features of CJD, although rarely, prolonged focal or generalized seizures can produce a similar imaging appearance.

Huntington’s disease (HD) ([Chap. 436](#)) is an autosomal dominant degenerative brain disorder. Clinical hallmarks of HD include chorea, behavioral disturbance, and executive impairment. Symptoms typically begin in the fourth or fifth decade of life, but there is a wide range, from childhood to >70 years. Memory is frequently not impaired until late in the disease, but attention, judgment, self-awareness, and executive functions are often deficient at an early stage. Depression, apathy, social withdrawal, irritability, and intermittent disinhibition are common. Delusions and obsessive-compulsive behavior may occur. Disease duration is variable but typically lasts ~15 years.

Normal-pressure hydrocephalus is a relatively uncommon but treatable syndrome. The clinical, physiologic, and neuroimaging characteristics of NPH must be carefully distinguished from those of other dementias associated with gait impairment. Historically, many patients treated for NPH have suffered from other dementias, particularly AD, vascular dementia, DLB, and progressive supranuclear palsy (PSP) ([Chap. 432](#)). For NPH, the clinical triad includes an abnormal gait (ataxic or apractic), dementia (usually mild to moderate, with an emphasis on executive impairment), and urinary urgency or incontinence. Neuroimaging reveals enlarged lateral ventricles (hydrocephalus) with little or no cortical atrophy, although the Sylvian fissures may appear propped open (so-called boxcarring), which can be mistaken for perisylvian atrophy. Crowding of dorsal frontal-parietal gyri helps distinguish NPH from other movement disorders, such as PSP and CBD, in which dorsal atrophy with sulcal widening is common. NPH is a communicating hydrocephalus with a patent aqueduct of Sylvius ([see Fig. 29-3](#)), in contrast to aqueductal stenosis, in which the aqueduct is small. Lumbar puncture opening pressure falls in the high-normal range, and the CSF protein, glucose, and cell counts are normal. NPH may be caused by obstruction to normal CSF flow over the cerebral convexities and delayed resorption into the venous system. The indolent nature of the process results in enlarged lateral ventricles with relatively little increase in CSF pressure. Presumed edema, stretching, and distortion of subfrontal white matter tracts may lead to clinical symptoms, but the precise underlying pathophysiology remains unclear. Some patients provide a history of conditions that produce meningeal scarring (blocking CSF resorption) such as previous meningitis, subarachnoid hemorrhage, or head trauma. Others with long-standing but asymptomatic congenital hydrocephalus may have adult-onset deterioration in gait or memory that is confused with NPH. In contrast to AD, the patient with NPH complains of an early and prominent gait disturbance without cortical atrophy on CT or MRI.

Numerous attempts have been undertaken to improve NPH diagnosis with various special studies and to predict the success of ventricular shunting. These tests include radionuclide cisternography (showing a delay in CSF absorption over the convexity) and various efforts to monitor and alter CSF flow dynamics, including a constant-pressure infusion test. None has proven to be specific or consistently useful. A transient improvement in gait or cognition may follow lumbar puncture (or serial punctures) with removal of 30–50 mL of CSF, but this finding has also not proved to be consistently predictive of postshunt improvement. Perhaps the most reliable strategy is a period of close inpatient evaluation before, during, and after lumbar CSF drainage. Occasionally, when a patient with AD presents with gait impairment (at times due to comorbid subfrontal vascular injury) and absent or only mild cortical atrophy on CT or MRI, distinguishing NPH from AD can be challenging. Hippocampal atrophy on MRI favors AD, whereas a characteristic “magnetic” gait with external hip rotation, low foot clearance, and short strides, along with prominent truncal sway or instability, favors NPH. The diagnosis of NPH should be avoided when hydrocephalus is not detected on imaging studies, even if the symptoms otherwise fit. Of those patients identified by careful diagnosis as having NPH, 30–50% will improve with ventricular shunting. Gait may improve more than cognition, but many reported failures to improve cognitively may have resulted from comorbid AD. Importantly, the presence of positive CSF AD biomarkers or amyloid PET is associated with lower likelihood of response to shunting. Short-lasting improvement is common. Patients should be carefully selected for shunting, because subdural hematoma, infection, and shunt failure are known complications and can be a cause for early nursing home placement in an elderly patient with previously mild dementia.

Intracranial hypotension, sometimes called sagging brain syndrome, is a disorder caused by low CSF pressure, leading to downward pressure on the subcortical structures and disruption of cerebral function. It presents in a variable manner with headache, often exacerbated by coughing or a Valsalva maneuver or by moving from lying to standing. Other common symptoms include dizziness, vomiting, disruption of sleep-wake cycles, and sometimes a progressive behavioral variant FTD-like syndrome ([Chap. 432](#)). Although sometimes idiopathic, this syndrome can be caused by CSF leaks secondary to lumbar puncture, head trauma, or spinal cord arachnoid cysts. Treatment consists of finding and patching the CSF leak.

Dementia can accompany *chronic alcoholism* ([Chap. 453](#)) and may result from associated malnutrition, especially of B vitamins, particularly [thiamine](#). Other poorly defined aspects of chronic alcoholism may, however, also produce cerebral damage. A rare idiopathic syndrome of dementia and seizures with degeneration of the corpus callosum has been reported primarily in male Italian red wine drinkers (Marchiafava-Bignami disease).

Thiamine (vitamin B₁) deficiency causes Wernicke’s encephalopathy ([Chap. 307](#)). The clinical presentation is usually a malnourished patient (frequently but not necessarily alcoholic) with confusion, ataxia, and diplopia resulting from inflammation and necrosis of periventricular midline structures, including dorsomedial thalamus, mammillary bodies, midline cerebellum, periaqueductal gray matter, and trochlear and abducens nuclei. Damage to the dorsomedial thalamus correlates most closely with the memory loss. Prompt administration of parenteral [thiamine](#) (100 mg intravenously for 3 days followed by daily oral dosage) may reverse the disease if given within the first days of symptom onset. Prolonged untreated [thiamine](#) deficiency can result in an irreversible and profound amnesic syndrome (Korsakoff’s syndrome) or even death.

In *Korsakoff’s syndrome*, the patient is unable to recall new information despite normal immediate memory, attention span, and level of consciousness. Memory for new events is seriously impaired, whereas knowledge acquired prior to the illness remains relatively intact. Patients are easily confused, disoriented, and cannot store information for more than a few minutes. Superficially, they may be conversant, engaging, and able to perform simple tasks and follow immediate commands. Confabulation is common, although not always present. There is no specific treatment because the previous [thiamine](#) deficiency has produced irreversible damage to the medial thalamic nuclei and mammillary bodies. Mammillary body atrophy may be visible on MRI in the chronic phase (see [Fig. 307-6](#)).

Vitamin B₁₂ deficiency, as can occur in pernicious anemia, causes a megaloblastic anemia and may also damage the nervous system ([Chaps. 99 and 442](#)). Neurologically, it most commonly produces a spinal cord syndrome (myelopathy) affecting the posterior columns (loss of vibration and position sense) and corticospinal tracts (hyperactive tendon reflexes with Babinski signs); it also damages peripheral nerves (neuropathy), resulting in sensory loss with depressed tendon reflexes. Damage to myelinated axons may also cause dementia. The mechanism of neurologic damage is unclear but may be related to a deficiency of S-adenosyl methionine (required for methylation of myelin phospholipids) due to reduced methionine synthase activity or accumulation of methylmalonate, homocysteine, and propionate, providing abnormal substrates for fatty acid synthesis in myelin. Use of histamine blockers or [metformin](#), vegan diets, autoimmunity against gastric parietal cells, and various causes of malabsorption are the typical causes for vitamin B₁₂ deficiency. The neurologic sequelae of vitamin B₁₂ deficiency may occur in the absence of hematologic manifestations, making it critical to avoid using the complete blood count (CBC) and blood smear as a substitute for measuring B₁₂ blood levels. Treatment with parenteral vitamin B₁₂ (1000 µg intramuscularly daily for a week, weekly for a month, and monthly for life for pernicious anemia) stops progression of the disease if instituted promptly, but complete reversal of advanced nervous system damage will not occur.

Deficiency of nicotinic acid (*pellagra*) is associated with skin rash over sun-exposed areas, glossitis, and angular stomatitis (**Chap. 333**). Severe dietary deficiency of nicotinic acid along with other B vitamins such as **pyridoxine** may result in spastic paraparesis, peripheral neuropathy, fatigue, irritability, and dementia. This syndrome has been seen in prisoners of war and in concentration camps but should be considered in any malnourished individual. Low serum folate levels appear to be a rough index of malnutrition, but isolated folate deficiency has not been proved as a specific cause of dementia.

CNS infections usually cause delirium and other acute neurologic syndromes. However, some chronic CNS infections, particularly those associated with chronic meningitis (**Chap. 139**), may produce a dementing illness. The possibility of chronic infectious meningitis should be suspected in patients presenting with a dementia or behavioral syndrome, who also have headache, meningismus, cranial neuropathy, and/or radiculopathy. Between 20–30% of patients in the advanced stages of HIV infection become demented (**Chap. 202**). Cardinal features include psychomotor retardation, apathy, and impaired memory. This syndrome may result from secondary opportunistic infections but can also be caused by direct infection of CNS neurons with HIV. Neurosyphilis (**Chap. 182**) was a common cause of dementia in the preantibiotic era; it is now uncommon but can still be encountered in patients with multiple sex partners, particularly among patients with HIV. Characteristic CSF changes consist of pleocytosis, increased protein, and a positive Venereal Disease Research Laboratory (VDRL) test.

Primary and metastatic *neoplasms of the CNS* (**Chap. 90**) usually produce focal neurologic findings and seizures rather than dementia, but if tumor growth begins in the frontal or temporal lobes, the initial manifestations may be memory loss or behavioral changes. An autoimmune, sometimes paraneoplastic, syndrome of dementia associated with occult carcinoma (often small-cell lung cancer) is termed *limbic encephalitis*. In this syndrome, confusion, agitation, seizures, poor memory, emotional changes, and frank dementia may occur. Paraneoplastic *encephalitis associated with N-methyl-D-aspartate (NMDA) receptor antibodies* presents as a progressive psychiatric disorder with memory loss and seizures; affected patients are often young women with ovarian teratoma. Autoimmune etiologies also include antibodies targeting *leucine-rich glioma-inactivated 1 (LGI1*; faciobrachial dystonic seizures); *contactin-associated protein-like 2 (Caspr2*; *insomnia, ataxia, myotonia*); and α -amino-3-hydroxy-5-methylisoxazole-4-propionic acid (AMPA)-receptor (limbic encephalitis with relapses), among others (**Chap. 94**).

A *nonconvulsive seizure disorder* (**Chap. 425**) may underlie a syndrome of confusion, clouding of consciousness, and garbled speech. Often, psychiatric disease is suspected, but an EEG demonstrates the epileptic nature of the illness. If recurrent or persistent, the condition may be termed *complex partial status epilepticus*. The cognitive disturbance often responds to anticonvulsant therapy. The etiology may be previous small strokes or head trauma; some cases are idiopathic. Nonconvulsive temporal lobe seizures can also emerge early in the course of AD.

It is important to recognize *systemic diseases* that indirectly affect the brain and produce chronic confusion or dementia. Such conditions include hypothyroidism; vasculitis; and hepatic, renal, or pulmonary disease. Hepatic encephalopathy may begin with irritability and confusion and slowly progress to agitation, lethargy, and coma.

Isolated vasculitis of the CNS (CNS granulomatous angiitis) (**Chaps. 363 and 427**) occasionally causes a chronic encephalopathy associated with confusion, disorientation, and clouding of consciousness. Headache is common, and strokes and cranial neuropathies may occur. Brain imaging studies may be normal or nonspecifically abnormal. CSF analysis reveals a mild pleocytosis or protein elevation. Cerebral angiography can show multifocal stenoses involving medium-caliber vessels, but some patients have only small-vessel disease that is not revealed on angiography. The angiographic appearance is not specific and may be mimicked by atherosclerosis, infection, or other causes of vascular disease. Brain or meningeal biopsy demonstrates endothelial cell proliferation and mononuclear infiltrates within blood vessel walls. The prognosis is often poor, although the disorder may remit spontaneously. Some patients respond to glucocorticoids or chemotherapy.

Chronic metal exposure represents a rare cause of dementia. The key to diagnosis is to elicit a history of exposure at work or home. Chronic lead poisoning from inadequately fire-glazed pottery has been reported. Fatigue, depression, and confusion may be associated with episodic abdominal pain and peripheral neuropathy. Gray lead lines appear in the gums, usually accompanied by an anemia with basophilic stippling of red blood cells. The clinical presentation can resemble that of acute intermittent porphyria (**Chap. 416**), including elevated levels of urine porphyrins as a result of the inhibition of δ -aminolevulinic acid dehydrase. The treatment is chelation therapy with agents such as ethylenediamine tetraacetic acid (EDTA). Chronic mercury poisoning produces dementia, peripheral neuropathy, ataxia, and tremulousness that may progress to a cerebellar intention tremor or choreoathetosis. The confusion and memory loss of chronic arsenic intoxication is also associated with nausea, weight loss, peripheral neuropathy, pigmentation and scaling of the skin, and transverse white lines of the fingernails (Mees' lines). Treatment is chelation therapy with **dimercaprol** (British anti-Lewisite, BAL). Aluminum poisoning is rare but was documented with the dialysis dementia syndrome, in which water used during renal dialysis was contaminated with excessive amounts of aluminum. This poisoning resulted in a progressive encephalopathy associated with confusion,

nonfluent aphasia, memory loss, agitation, and, later, lethargy and stupor. Speech arrest and myoclonic jerks were common and associated with severe and generalized EEG changes. The condition has been eliminated by the use of deionized water for dialysis.

Recurrent head trauma in professional athletes may lead to a dementia previously referred to as “punch-drunk” syndrome or *dementia pugilistica* but now known as chronic traumatic encephalopathy (CTE) to signify its relevance to contact sport athletes other than boxers (**Chap. 443**). The symptoms can be progressive, beginning late in an athlete’s career or, more often, after retirement. Early in the course, a personality change occurs, associated with social instability, explosive rage, and sometimes paranoia and delusions. Later, memory loss progresses to full-blown dementia, often associated with parkinsonian signs and ataxia or intention tremor. At autopsy, the cerebral cortex shows tau-immunoreactive NFTs that are more prominent than amyloid plaques (which are usually diffuse or absent rather than neuritic). NFTs and tau-positive reactive astrocytes are often clustered in the depths of cortical sulci and in a perivascular distribution. TDP-43 inclusions (**Chap. 432**) have also been reported, highlighting the overlap with the FTD spectrum. Loss of neurons in the substantia nigra is a variable feature, and some with TDP-43 inclusions also develop motor neuron disease (MND) (**Chap. 437**).

Chronic subdural hematoma (**Chap. 443**) is also occasionally associated with dementia, often in the context of underlying cortical atrophy from conditions such as AD or HD.

Transient global amnesia (TGA) is characterized by the sudden onset of a severe episodic memory deficit, usually occurring in persons aged >50 years. Often the amnesia occurs in the setting of an emotional stimulus or physical exertion. During the attack, the individual is alert and communicative, general cognition seems intact, and there are no other neurologic signs or symptoms. The patient may seem confused and repeatedly ask about his or her location in place and time. The ability to form new memories returns after a period of hours, and the individual returns to normal with no recall for the period of the attack. Frequently no cause is determined, but cerebrovascular disease, epilepsy (7% in one study), migraine, or cardiac arrhythmias have all been implicated. Approximately one-quarter of patients experience recurrent attacks. Rare instances of permanent memory loss have been reported in patients with TGA-like spells, usually representing ischemic infarction of the hippocampus or dorsomedial thalamic nucleus bilaterally. Seizure activity due to AD should always be suspected in this syndrome.

The *ALS/parkinsonian/dementia complex of Guam* is a rare degenerative disease that has occurred in the Chamorro natives on the island of Guam. Individuals may have any combination of parkinsonian features, dementia, and MND. The most characteristic pathologic features are the presence of NFTs in degenerating neurons of the cortex and substantia nigra and loss of motor neurons in the spinal cord, although recent reanalysis has shown that some patients with this illness also show coexisting TDP-43 pathology. Epidemiologic evidence supports a possible environmental cause, such as exposure to a neurotoxin or an infectious agent with a long latency period. One interesting but unproven candidate neurotoxin is the seed of the false palm tree, which Guamanians traditionally used to make flour. The amyotrophic lateral sclerosis (ALS) syndrome is no longer present in Guam, but a dementing illness with rigidity continues to be seen.

Rarely, adult-onset leukodystrophies, lysosomal-storage diseases, and other genetic disorders can present as a dementia in middle to late life. Metachromatic leukodystrophy (MLD) causes a progressive psychiatric or dementia syndrome associated with an extensive, confluent frontal white matter abnormality. MLD is diagnosed by measuring reduced arylsulfatase A enzyme activity in peripheral white blood cells. Adult-onset presentations of adrenoleukodystrophy have been reported in female carriers, and these patients often feature spinal cord and posterior white matter involvement. Adrenoleukodystrophy is diagnosed by demonstrating increased levels of plasma very-long-chain fatty acids. CADASIL is another genetic syndrome associated with white matter disease, often frontally and temporally predominant. Diagnosis is made with skin biopsy, which shows osmophilic granules in arterioles, or increasingly through genetic testing for mutations in Notch 3. The neuronal ceroid lipofuscinoses are a genetically heterogeneous group of disorders associated with myoclonus, seizures, vision loss, and progressive dementia. Diagnosis is made by finding eosinophilic curvilinear inclusions within white blood cells or neuronal tissue.

Psychogenic amnesia for personally important memories can be seen. Whether this results from deliberate avoidance of unpleasant memories, outright malingering, or unconscious repression remains unknown and probably depends on the patient. Event-specific amnesia is more likely to occur after violent crimes such as homicide of a close relative or friend or sexual abuse. It may develop in association with severe drug or alcohol intoxication and sometimes with schizophrenia. More prolonged psychogenic amnesia occurs in fugue states that also commonly follow severe emotional stress. The patient with a fugue state suffers from a sudden loss of personal identity and may be found wandering far from home. *In contrast to neurologic amnesia, fugue states are associated with amnesia for personal identity and events closely associated with the personal past.* At the same time, memory for other recent events and the ability to learn and use new information are preserved. The episodes usually last hours or days and occasionally weeks or months while the patient takes on a new identity. On recovery, there is a residual amnesia gap for the period of the fugue.

Very rarely does selective loss of autobiographic information reflect a focal injury to the brain areas involved with these functions.

Psychiatric diseases may mimic dementia. Severely depressed or anxious individuals may appear demented, a phenomenon sometimes called *pseudodementia*. Memory and language are usually intact when carefully tested, and a significant memory disturbance usually suggests an underlying dementia, even if the patient is depressed. Patients in this condition may feel confused and unable to accomplish routine tasks. Vegetative symptoms, such as insomnia, lack of energy, poor appetite, and concern with bowel function, are common. Onset is often more abrupt, and the psychosocial milieu may suggest prominent reasons for depression. Such patients respond to treatment of the underlying psychiatric illness. Schizophrenia is usually not difficult to distinguish from dementia, but occasionally the distinction can be problematic. Schizophrenia generally has a much earlier age of onset (second and third decades of life) than most dementing illnesses and is associated with intact memory. The delusions and hallucinations of schizophrenia are usually more complex, bizarre, and threatening than those of dementia. Some chronic schizophrenics develop an unexplained progressive dementia late in life that is not related to AD. Conversely, FTD, HD, vascular dementia, DLB, AD, or leukoencephalopathy can begin with schizophrenia-like features, leading to the misdiagnosis of a psychiatric condition. Later age of onset, significant deficits on cognitive testing, or the presence of abnormal neuroimaging suggest a degenerative condition. Memory loss may also be part of a *conversion disorder*. In this situation, patients commonly complain bitterly of memory loss, but careful cognitive testing either does not confirm the deficits or demonstrates inconsistent or unusual patterns of cognitive problems. The patient's behavior and "wrong" answers to questions often indicate that he or she understands the question and knows the correct answer.

Clouding of cognition by *chronic drug or medication use*, often prescribed by physicians, is an important cause of dementia. Sedatives, tranquilizers, and analgesics used to treat insomnia, pain, anxiety, or agitation may cause confusion, memory loss, and lethargy, especially in the elderly. Discontinuation of the offending medication often improves mentation.

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